Efficient Agreement Detection for Online Discourse

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Background and motivation for the project:

We started our project by deciding we wanted to work on a sentiment analysis project, but sentiment analysis by itself seemed extensively done, and limited in actual application. Thus, we instead pivoted to agreement detection, a subset of sentiment analysis which we thought was more valuable for analysing online discourse. Whereas sentiment analysis only evaluates the tone of a particular message in isolation, allowing for experts to approximately gauge the opinions of users towards a certain idea, with agreement detection one can directly ascertain the same information, as well as account for context, which allows for analysis of more nuanced discussions with higher accuracy.

Of course, it is possible to perform agreement analysis using an LLM. However, when analysing particularly large amounts of data using consumer-grade hardware, they are particularly inefficient. This project is our attempt to devise a way to perform agreement detection on large amounts of data efficiently, while retaining approximately the performance of an LLM.

Related prior work:

While sentiment analysis has many real world examples and a rich collection of applications, discourse analysis has received less attention. The Multi-Genre Natural Language interference (MultiNLI) is the closest to our model in scope and purpose, but is more broad then our agreement/non agreement model.

MultiNLI focuses on the logical entailment of the sentences, and how two independent sentences, the premise and the hypothesis, relate to one another. This relation could be one of neutrality, of contradiction or of entailment. The logic inherent in how an idea is expressed and whether that idea is reinforced or contradicted runs parallel to our own logic of agreement and disagreement. In our research for this project, we found MultiNLI to be an interesting tangent, but not a direct inspiration. Since the dataset we chose to work from was directly sourced from reddit, and therefore included informal language and complex opinions, our model needed to more directed towards that type of agreement/disagreement.

Statement of objectives:

Going into the project, we sought to build a model that could quickly, and with acceptable accuracy detect whether a **response** statement was in agreement with a

premise statement. Determining this could have a multitude of applications, but mainly to determine the level of discourse (the amount of fighting) going on in any particular forum or social media thread.

We estimated that a large language model could easily compute agreement between pairs of responses and premises with high accuracy owing to the complexity of its natural language processing. However, the process would be slow, and further would be intensive on GPU resources. Furthermore, the payment barrier of most commercially available LLMs, and their limitations on data processing make it difficult to do this, given that some threads or forums could have thousands of comments.

Our objective is to build a model that will fulfill our goals orders of magnitude faster and less intensive than an LLM, though we do not expect our model to be as accurate as an LLM. Further, by training specifically on sets of reddit comments our model will be better suited to handling the intricacies of online debate.

Methods:

To achieve a comparable level of accuracy to an LLM while being significantly faster, we decided to take advantage of transformer models. To provide a brief summary as to what a transformer model is, they are effectively models which can take in some input, like say a word or sentence, and as an output provide a high-dimensional vector composed of various features corresponding to that input, called an embedding. These embeddings allow us to do various things, like for example compare two words by finding the distance between their embeddings.

While LLMs also use transformer models, they are paired with more elements to allow the model to actually provide a response. We decided to use transformer models--specifically sentence transformers, which create fixed length embeddings which represent sentences, given a string input--as a means to extract the features in a sentence which are relevant to agreement and disagreement, before providing these features to a feedforward neural network for classification.

Since there is only a classification network beyond the transformer model, this allows our agreement disagreement model to be significantly smaller than an LLM. At the absolute largest in our development process, our model never exceeded a total ~300 million parameters, while even an LLM which is considered small would easily clear 3 and a half billion. To summarize, our model is mostly an encoder-only model, whereas an LLM is an encoder and decoder architecture, making it significantly lighter and faster.

Of course, there is a limited number of features which can be fit into a vector. The sentence transformers we used could not have possibly provided the relevant features of a sentence for all possible domains, and almost all pertained sentence transformers--which we used because it would have been impossible to create a sentence transformer from scratch with our resources--provide features only for a general set of areas. These include features that describe the semantic meaning of a sentence for tasks like finding sentences that mean similar things, or the features that make sentences that imply other sentences closer together in the feature space for natural language inference tasks, but nothing as specific as features for agreement and disagreement. Thus, we had to fine tune a general purpose sentence transformer to achieve acceptable efficiency for our downstream task for agreement detection.

To do this, we used the pretrained model all-mpnet-base-v2, and fine-tuned it with contrastive loss by setting all ‘agreement’ labels in our dataset to 1, and all ‘disagreement’ labels in our dataset to 0. Effectively, this served to move the embeddings of sentences that don’t agree with each other farther in the vector space, while moving those that agree with each other closer--making the features within those embeddings more relevant to agreement disagreement overall.

Once the sentence transformer was fine-tuned, we could then train our neural network to interpret those embeddings, and provide the output that we desire. For the network, we used cross entropy loss and softmax, as it was responsible for a binary classification task.

Dataset

The dataset we used was DEBAGREEMENT, an open source agreement/disagreement dataset of reddit interactions developed by researchers at Oxford University in 2021. The dataset contains comment/reply interactions from subreddits such as r/Democrats, and r/Republicans, and contains interactions relating to US politics, Brexit, climate change and the Black Lives Matter protests. Each comment/reply pair is labelled with whether the reply agrees, disagrees or is neutral towards the initial comment. DEBAGREEMENT contains over 40,000 comment/reply pairs.

Validation

Due to the sheer amount of data we had access to via DEBAGREEMENT, we decided to split the data into 95% training, and 5% validation. Even though 5% seems particularly small, the sheer size of the dataset meant that this was still thousands of sentence-pairs to test our model against.

Qualitative Results

Along with our model, we implemented a front end which utilizes a scraper to collect premise-response pairs from a provided link, and then produces a table showing all the pairs along with their associated predictions.

Thread 1 URL

	Index	Text A	Text B
1	Agree	America blows up...	Amen dude, you literally summed it up, he's doing a ...
2	Agree	Amen dude, you literally summed it up, he's doing a ...	I agree. Funny man no doubt and pretty damn solid ...
3	Agree	Trump is like a circus clown, he is all fun and giggles, ...	America blows up...
4	Agree	America blows up...	Amen dude, you literally summed it up, he's doing a ...
5	Agree	Amen dude, you literally summed it up, he's doing a ...	...
6	Agree	Trump is like a circus clown, he is all fun and giggles, ...	America blows up...
7	Agree	America blows up...	...
8	Not Agree	Trump is like a circus clown, he is all fun and giggles, ...	but once he has power, he becomes the Joker....
9	Not Agree	but once he has power, he becomes the Joker....	He hasn't really done much honestly. Just a lot of ...
10	Not Agree	He hasn't really done much honestly. Just a lot of ...	I mean it hasn't just survived, but it did well. The most...
11	Not Agree	I mean it hasn't just survived, but it did well. The most...	Reading these comments restores hope their are ...
12	Not Agree	Trump is like a circus clown, he is all fun and giggles, ...	but once he has power, he becomes the Joker....
13	Not Agree	but once he has power, he becomes the Joker....	He hasn't really done much honestly. Just a lot of ...
14	Not Agree	He hasn't really done much honestly. Just a lot of ...	I mean it hasn't just survived, but it did well. The most...
15	Not Agree	I mean it hasn't just survived, but it did well. The most...	...
16	Not Agree	Trump is like a circus clown, he is all fun and giggles, ...	but once he has power, he becomes the Joker....
17	Not Agree	but once he has power, he becomes the Joker....	He hasn't really done much honestly. Just a lot of ...
18	Not Agree	He hasn't really done much honestly. Just a lot of ...	...

Thread 2 URL

	Index	Text A	Text B
1	Not Agree	Empathetic people are called normal people....	You'd be surprised at how small the population of ...
2	Not Agree	Correct me if I'm wrong, but isn't an empath just ...	Empathetic people are called normal people....
3	Agree	Empathetic people are called normal people....	Fair enough
4	Not Agree	Correct me if I'm wrong, but isn't an empath just ...	Empathetic people are called normal people....
5	Agree	Empathetic people are called normal people....	Emotional dysregulation...
6	Not Agree	Correct me if I'm wrong, but isn't an empath just ...	Empathetic people are called normal people....
7	Not Agree	Empathetic people are called normal people....	But there is a range of empathetic ability and "normal...
8	Agree	Correct me if I'm wrong, but isn't an empath just ...	You're correct. Empaths are just empathetic people, ...
9	Not Agree	Sometimes an opinion is unpopular because it is ...	Not sometimes, actually very often.
10	Agree	you're right and I think people are missing the point. ...	Upvote for the last sentence. Those are narcissists wh...
11	Agree	I feel like an important distinction should be made ...	Yeah, one of my co workers says this and how nice sh...
12	Not Agree	I feel like an important distinction should be made ...	I find the people who tell everyone they are empaths ...
13	Not Agree	When is this empath/narcissist fashion scheduled to ...	Never not with social media
14	Agree	Never not with social media	Never as long as people with low to middling IQs exis...
15	Agree	Never as long as people with low to middling IQs exis...	The "I crap on dumb people to give off the impressio...
16	Not Agree	When is this empath/narcissist fashion scheduled to ...	Never not with social media
17	Agree	Never not with social media	Never as long as people with low to middling IQs exis...
18	Agree	Never as long as people with low to middling IQs exis...	...
19	Not Agree	When is this empath/narcissist fashion scheduled to	Never not with social media

(Only the first 20 for both are displayed, as we do not have the space to cover the combined 120+ pairs for both threads. Unfortunately we couldn't submit CSVs along with this assignment)

As a general observation, our model performed quite well when commenters were agreeing or disagreeing, but less well if they were neutral or noncommittal. Sarcasm, rare slang, and nonpolitical discussion also seems to reduce performance. However, as a whole for most agreement-disagreement, it works well, and can effectively capture the overall 'trend' of a particular thread.

Quantitative

Our model compared to LLMs on an NVIDIA GeForce GTX 1650 GPU. Both LLMs are 4 bit quantized and are being run locally:

Measures of	Our model (300	Phi 3 (3.5 billion	Mistral 7b (7 billion

performance	million params)	params)	params)
Validation accuracy	~80%	Estimate 85%	Estimate 90%
Inference Speed	58 ms / pair	624 ms / pair	1448 ms / pair

We could not gauge the accuracy of Phi 3 and Mistral 7b on our validation set, due to us not having the time to create a system to programmatically feed in all the prompts. We instead asked chat GPT 4, a larger model, its estimated accuracy for sentiment analysis, which is a simpler task than agreement detection, to which it responded 85% to 90%, which we used for our estimates above. Considering both Phi 3 and Mistral 7b are significantly smaller than chat GPT 4, and that we asked for its estimates for an easier task, we believe that these are fair estimates as we are almost certainly overestimating the LLMs we are comparing our model against.

Note that the data in this table was acquired from a computer with different hardware than the one we used to collect our data for our presentation, so some figures will be different. However, it still works for comparison purposes between the items present in this table.

The above LLM inference speeds were collected by using ollama --verbose. We provided them the following sentence pairs to classify and took their average inference speed. These are the LLM evaluation speeds, meaning that we are not including the time it took for the LLM to actually respond. We ran each of these sentences 3 times for each LLM and took the fastest. We also specifically chose short, easy sentences to get a highball of their speed.

Sentences	Phi-3 eval speed	Mistral 7b eval speed
“Donald trump is great,” “I used to think he was trash, but now I see he will save the country,”	0.915972s	1.442065s
“Hey I really like the weather today” “It kinda cold though”	0.538535s	1.319768s

“Aren’t you still working on the 3106 final project?” “Nah I’m done.”	0.416636s	1.583569s

Limitations of the work and directions for future work

Utilising all the methods outlined above we were able to attain a validation accuracy of roughly 80%. As mentioned above, our mislabeled sets are mostly ones containing heavy sarcasm, rare slang, or ideas that do not relate to politics, and more neutral, non-committal responses. This fits with our dataset being composed of political opinions and reactions, which correspond to a relatively consistent “political debate” linguistic pattern picked up by our SBERT vectorization model. In further testing with our reddit scraper program we found that the farther debate topics got from politics the more prone to mistakes our model became. For instance, testing on threads from r/nba which we validated with ChatGPT found a 68% validation accuracy, well below the 80% we achieved with DEBAGREEMENT. This could be chalked up to the lack of ability to pick up on never-before-seen NBA terminology and debate patterns. Future directions for this project would be further diversifying our dataset, possibly using classifications generated from LLMs.

Another limitation of our model is its simple Agreement/Non-Agreement classification. It must be explicitly stated that our model does not distinguish between disagreement and neutrality from the response to the premise. Though DEBAGREEMENT has 3 classes, agree, disagree and neutral, we combined the latter two to simplify the problem into that of binary classification. Future directions for this project would be to integrate neutrality. This might be accomplished by making a second model which classifies disagreement vs non disagreement, then using fuzzy set logic and probabilities to detect neutrality in statements.

Implications of the work

Our model and any future version of it could have significant implications on the study and analysis of online debate. As polarization, political and otherwise, is generally increasing both in real life and particularly online, the ability to monitor and assess the level of disagreement and discourse on public forums is an excellent way to gauge public opinion and political divisiveness.

Inversely, forums and communities that display unusually high levels of agreement are typically more biased and may be silencing dissenting opinions and discouraging honest

debate. As many people receive their news on the internet, it is important that they be made aware of these communities and consider the level of discourse on a forum before blindly believing what they see.

References / Resources Used:

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